Motion in a Central Potential — approved segments

24 approved segments, in reading order

CHAPTER 3

MOTION IN CENTRAL POTENTIAL $V(\vec{r})$

$\vec{L} = \mu r^2 \dot{\phi} \hat{z}$ $\quad \vec{r} = r\hat{r}$ $\quad \vec{p} = \mu \dot{\vec{r}}$

DI

ℓ

page01_seg01.png

MOTION IN CENTRAL POTENTIAL $V(r)$

$$\vec{L} = \mu r^2 \dot{\phi} \hat{z} \qquad \vec{r} = r\hat{r} \qquad \vec{p} = \mu \dot{\vec{r}}$$

CONST OF MOTION

$$L = T - V = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - V(r)$$

$$\phi \text{ CYCLIC} \Leftrightarrow \frac{\partial L}{\partial \phi} = 0 \Rightarrow p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \mu r^2 \dot{\phi} \equiv \ell$$

CONSERVED

$$V_{eff}(r) = V(r) + \frac{\ell^2}{2\mu r^2}$$

$$E = T + V = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) + V(r) = \frac{\mu}{2}\dot{r}^2 + V_{eff}(r)$$

ℓ

page01_seg02.png

$$E = T + V = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) + V(r) = \frac{\mu}{2}\dot{r}^2 + V_{eff}(r)$$

UNIVERSAL METHOD ⇒ COMPLETE SOLUTION

$$t = \int^{r} \frac{dr'}{\sqrt{\frac{2}{\ }\left(E - V(r') - \ell^2 \dots\right)}} \qquad t = t(r)$$

$\mu\ddot{r} =$

page01_seg03.png

$$t = \int_{r_0}^{r} \frac{dr'}{\sqrt{\frac{2}{\mu}\left(E - V(r') - \frac{\ell^2}{2\mu r'^2}\right)}}$$

$$t = t(r)$$

$$\phi = \phi_0 + \int_{t_0}^{t} \frac{\ell}{\mu r^2(t')}\, dt'$$

$$\phi = \phi(t)$$

$\mu \ddot{r} =$

$\frac{\partial}{\partial r}($

LHS:

page01_seg04.png

DIFF EQN FOR $r = r(\phi)$

$$\ell = \mu r^2 \dot{\phi} \Rightarrow \dot{\phi} = \frac{\ell}{\mu r^2} = \frac{d\phi}{dt}$$

$dF \quad dF \; d\phi \; = \; \ell \; dF$

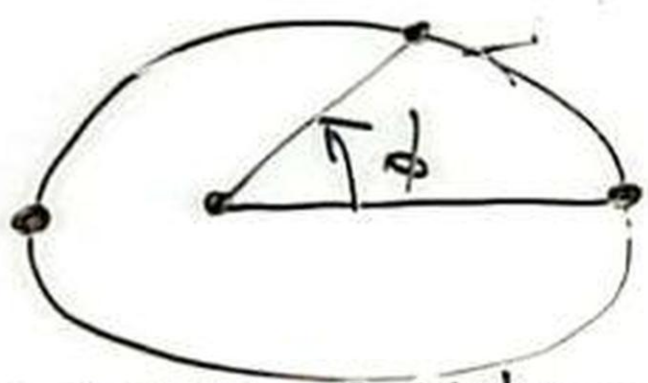
page01_seg05.png

$$\frac{dF}{dt} = \frac{dF}{d\phi}\frac{d\phi}{dt} = \underbrace{\frac{l}{\mu r^2}\frac{dF}{d\phi}}$$

$\phi \equiv l$

$F(1/u) = F($

page01_seg06.png



$$-\frac{\partial V}{\partial r} = F(r)$$

Radial Eq of Motion

$$\mu \ddot{r} = -\frac{\partial V_{eff}}{\partial r} = -\frac{\partial}{\partial r}\left(V(r) + \frac{\ell^2}{2\mu r^2}\right)$$

page01_seg07.png

$= r(\phi)$

$\frac{d\phi}{dt}$

$- \frac{dF}{d\phi}$

$$F(r) = -\frac{\partial V}{\partial r}$$

$$\frac{dV}{dr} = \frac{dV}{du}\frac{du}{dr} = -u^2 \frac{dV}{du}$$

$$\frac{du}{dr} = -\frac{1}{r^2} = -u^2$$

$$F(1/u) = F(r) = -\frac{dV}{dr} = u^2 \frac{dV(1/u)}{du}$$

GRAVITATI

KEP

$$V = -\frac{k}{r} = -k$$

$$\boxed{\frac{d^2u}{d\phi^2} + u =}$$

page01_seg08.png

$$-\frac{\partial V}{\partial r} = F(r)$$

$$F(1/u) = F(r) = -\frac{dV}{dr} = u^2 \frac{dV(1/u)}{du}$$

$$-F(1/u) = -u^2\, dV(1/u)/du$$

$$-F\left(\frac{1}{u}\right) = \frac{l^2}{\mu} u^2 \left[u + \frac{d^2u}{d\phi^2} \right]$$

ASIDE

$$\tilde{u} = u - \frac{k\mu}{l^2}$$

$$\boxed{\frac{d^2\tilde{u}}{d\phi^2} + \tilde{u} = 0}$$

page01_seg09.png

$$V(r) + \frac{\ell^2}{2\mu r^2}\Big)$$

$$-\cancel{u^2}\,\frac{dV}{du} = \frac{\ell^2}{\mu}\cancel{u^2}\left[u + \frac{d^2u}{d\phi^2}\right]$$

$$\boxed{\frac{d^2u}{d\phi^2} + u = 0}$$

$$u(\phi) = k\mu$$

page01_seg10.png

$)$ $\qquad \mu\ddot{r} = F(r) + \frac{\ell^2}{\mu r^3} \qquad u = \frac{1}{r} \qquad \frac{du}{d\phi} = \frac{du}{dr}\frac{dr}{d\phi} = \frac{-1}{r^2}\frac{dr}{d\phi} \qquad \frac{d^2u}{d\phi^2}$

(t) $\qquad \frac{\partial}{\partial r}\left(r^{-2}\right) = (-2)\, r^{-3} \qquad \frac{\ell^2}{\mu r^3} = \frac{\ell^2}{\mu} u^3 \Rightarrow \text{RHS} : F(1/u) + \frac{\ell^2}{\mu} u^3$

EQ

page01_seg11.png

$$\underline{\text{LHS}}: \ \mu \frac{d}{dt}\frac{dr}{dt} = \mu \frac{d}{dt}\frac{l}{\mu r^2}\frac{dr}{d\phi} = \cancel{\mu}\frac{l}{\cancel{\mu} r^2}\frac{d}{d\phi}\left(\frac{l}{\mu r^2}\frac{dr}{d\phi}\right)$$

page01_seg12.png

LHS:

$$\mu \ddot{r} = -\frac{\ell^2}{\mu}\frac{1}{r^2}\frac{d}{d\phi}\frac{du}{d\phi} = -\frac{\ell^2}{\mu}u^2\frac{d^2u}{d\phi^2}$$

page01_seg13.png

LHS = RHS

$$\boxed{-\frac{\ell^2}{\mu} u^2 \frac{d^2 u}{d\phi^2} = F(1/u) + \frac{\ell^2}{\mu} u^3}$$

GENERIC EQ. OF TRAJECTORY FOR ARBITRARY $V(r)$

page01_seg14.png

$$\boxed{\frac{d^2u}{d\phi^2} + u = -\frac{\mu}{\ell^2}\frac{d}{du}V(1/u)}$$

$u(\phi) = \frac{k\mu}{\ell^2}$

EQ FOR TRAJECTORY $r(\phi)$

SOLVE FOR GIVEN

$u = u_0$ FOR $\phi = \phi_0$ $\frac{du}{d\phi} = 0$

CTORY

$\frac{du}{d\phi} = 0 \Leftrightarrow$ MAX OR MIN r POINT

page01_seg15.png

GRAVITATION CASE

KEPLER PROBLEM

$$V = -\frac{k}{r} = -ku \qquad \frac{dV}{du} = -k$$

d^2u … HARMONIC

page01_seg16.png

$$\boxed{\frac{d^2 u}{d\phi^2} + u = \frac{k\mu}{\ell^2}}$$

HARMONIC OSCILLATOR WITH SHIFTED ORIGIN

ASIDE

$$\tilde{u} = u - \frac{k\mu}{\ell^2}$$

$$\boxed{\frac{d^2\tilde{u}}{d\phi^2} + \tilde{u} = 0}$$

HOMOGENEOUS SOLUTION

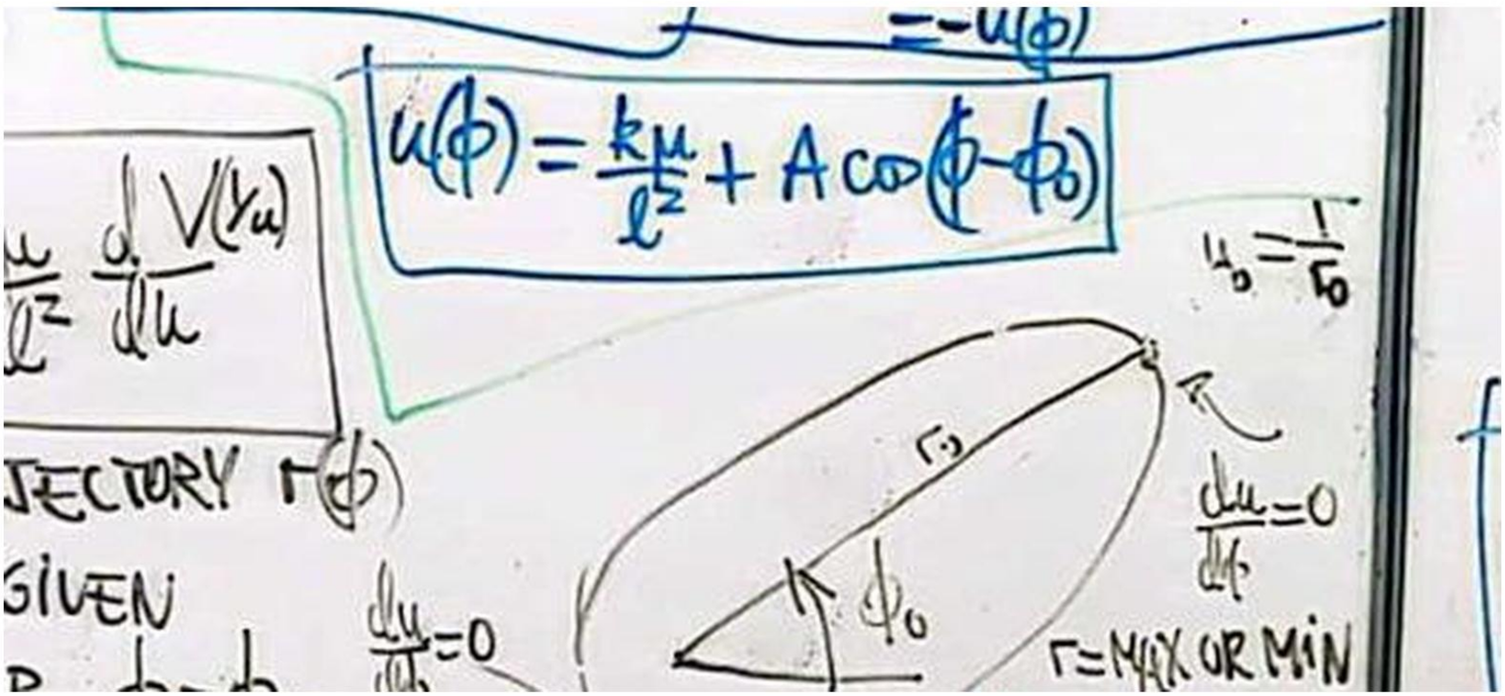
$$\tilde{u}(\phi) = A\cos(\phi - \phi_0)$$

$$\tilde{u}'(\phi) = -A\sin(\phi - \phi_0)$$

$$\tilde{u}''(\phi) = -A\cos(\phi - \phi_0) = -\tilde{u}(\phi)$$

$$\boxed{u(\phi) = \frac{k\mu}{\ell^2} + A\cos(\phi - \phi_0)}$$

page01_seg17.png



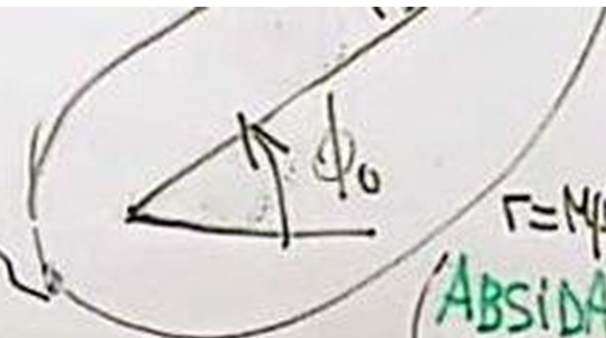
page01_seg18.png

JECTORY $r(\phi)$

GIVEN

R $\phi = \phi_0$ $\frac{du}{d\phi} = 0$

$\Rightarrow$ MAX OR MIN r POINT

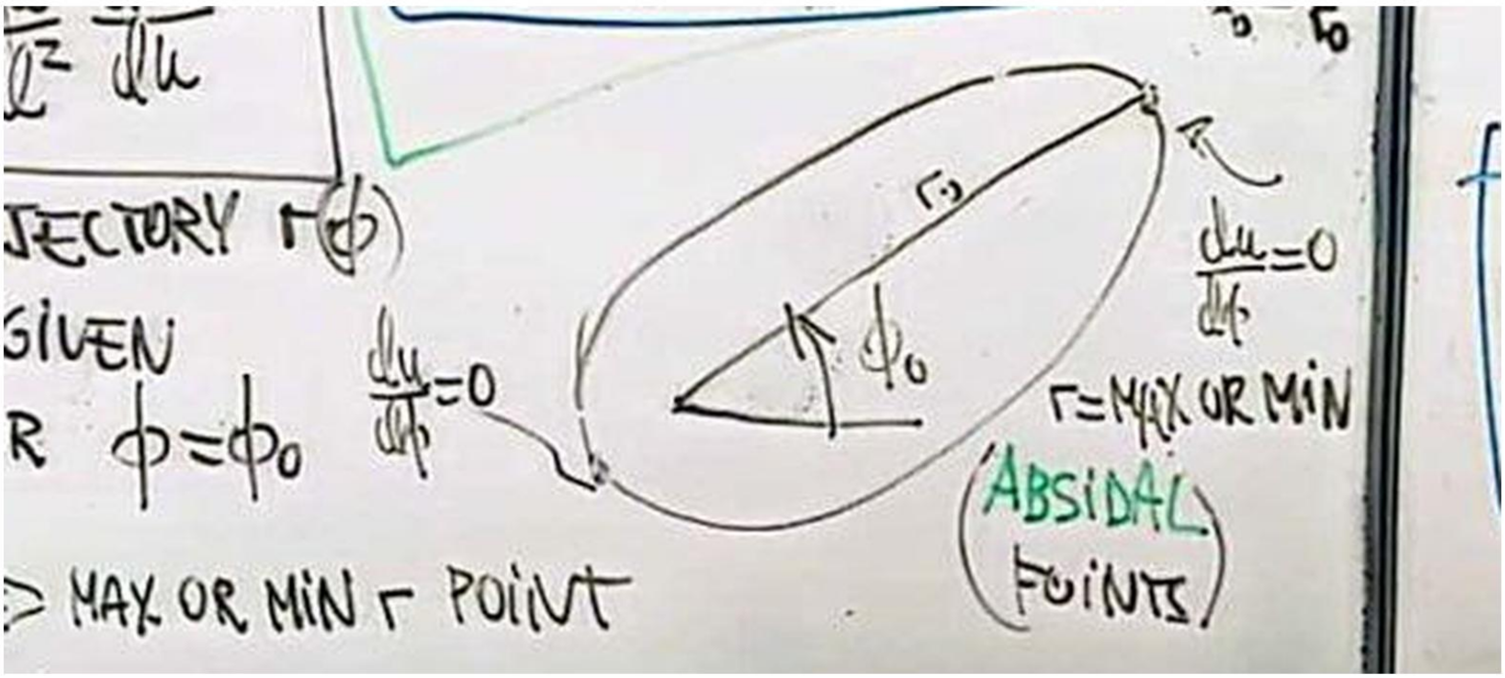


$\frac{du}{d\phi} = 0$

r = MAX OR MIN

(ABSIDAL POINTS)

page01_seg19.png



page01_seg20.png

$$\boxed{u(\phi) = \frac{k\mu}{\ell^2}\left[e\cos(\phi-\phi_0)+1\right] = \frac{1}{r}}$$

e CONSTANT $\quad e \equiv \dfrac{A}{k\mu/\ell^2}$

USING THE UNIVERSAL METHOD

$$u'' = -u + \frac{k\mu}{\ell^2}$$

page01_seg21.png

$$u'' = -u + \frac{R\mu}{\ell^2}$$

$$\Downarrow \times u'(\phi)$$

$$\frac{du}{} = (u')^2 = \sqrt{\cdot \quad -}$$

page01_seg22.png

$$\frac{du}{d\phi} = (u')^2 = \sqrt{\cdot \quad -}$$

$$d\phi = \frac{du}{u'}$$

$$\vdots$$

page01_seg23.png

$$\boxed{u = \frac{\mu k}{\ell^2}\left[1+\sqrt{1+\frac{2\ell^2 E}{\mu k^2}}\,\cos(\phi-\phi_0)\right] = \frac{1}{r}}$$

$$\boxed{e \equiv \sqrt{1+\frac{2\ell^2 E}{\mu k^2}}}$$

page01_seg24.png